Project Title: Personalised Offer Recommendation
Intern Name: Alina Lakra
Organization: WiseAnalytics
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Module Description

The project aims to develop an artificial intelligence-based personalization system in a retail industry where customers are offered specific discounts and offers. This system will use the history of customers' purchases, simulated price competition from competitors, and trend information about products to predict the probability of a customer accepting a specific offer.

One of the main features of this system is its ability to provide explanations for specific recommendations. This helps businesses understand the reasons behind a specific offer being recommended by the system.

The system utilizes machine learning algorithms to analyze customers' behavior patterns and provide personalization marketing strategies.

Input File Types

The system uses structured datasets in the following format:

- Primary Format: CSV file
- Optional Formats: JSON, TXT

Dataset Features:

- customer_id – Unique identifier
- product_category – Product category (e.g., Shoes, Clothing)
- price – Product price
- discount_offered – Discount percentage
- trend_score – Popularity/trend indicator
- competitor_price – Competitor pricing
- purchased – Target variable (1 = accepted, 0 = not accepted)

Methodology / Features

1. **Exploratory Data Analysis (EDA):**
   - Data inspection and summary statistics
   - Identification of missing values
   - Understanding feature distributions
   - Basic correlation analysis

2. **Data Preprocessing:**
   - Selection of relevant numerical features
   - Handling missing/null values
   - Preparing feature matrix (X) and target variable (y)

3. **Feature Engineering:**
   - **Inclusion of behavioural and pricing features**
   - Capturing influence of:
     -
     - Market trends
     - Competitor prices

4. **Model Building:**
   - A Random Forest Classifier is used
   - The model is trained to predict whether an offer will be accepted
   - Dataset is split into training and testing sets

5. **Model Evaluation:**
   - **Accuracy and prediction consistency are checked**
   - Model performance is validated using test data

6. **Explainability (SHAP):**
   - SHAP (SHapley Additive exPlanations) is applied
   - It explains how each feature contributes to predictions
   - Helps answer:
   - Why a customer received a specific offer
   - Which factors influenced the decision most

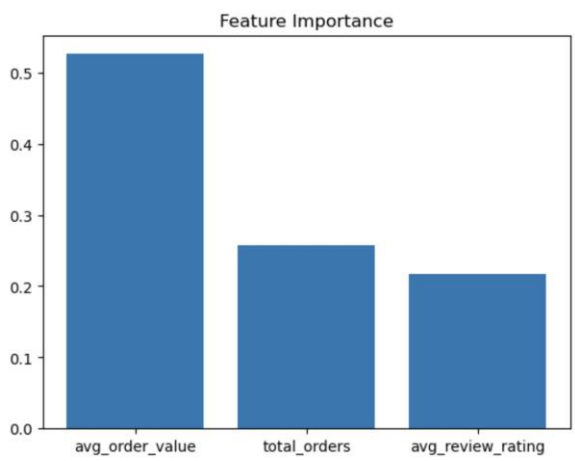
Flow Diagram

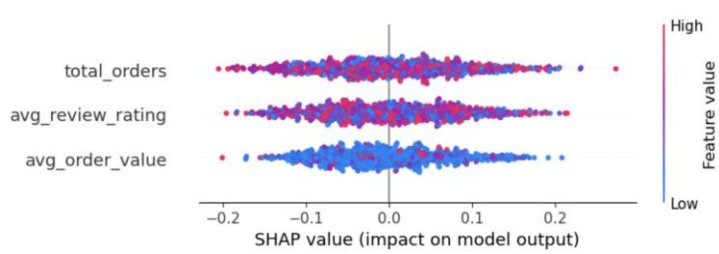
Input Dataset → EDA → Data Preprocessing → Feature Engineering

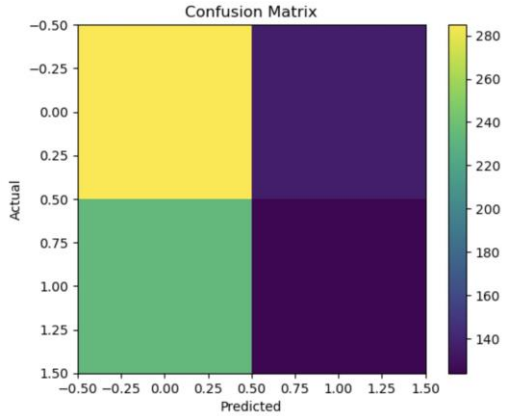
→ Model Training → Prediction → SHAP Explanation → Output

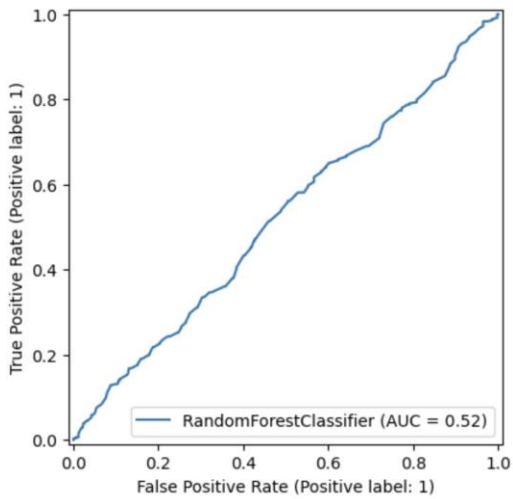
Visualization & Insights

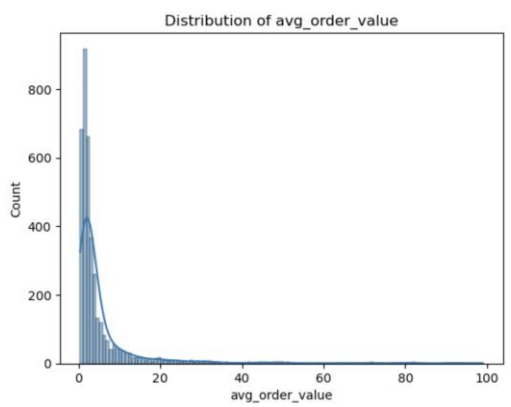
- Feature importance analysis shows:
  - Trend score and discount have strong impact
  - Competitor price influences decision behavior
- SHAP plots highlight:
  - Contribution of each feature to prediction
  - Positive and negative influences
- Observations:
  - Higher discounts increase acceptance probability
  - Trending products are more likely to be purchased
  - Competitive pricing improves conversion

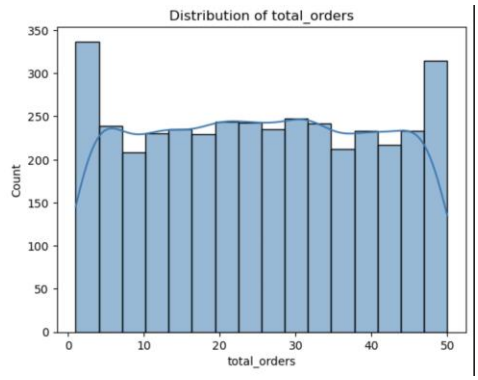


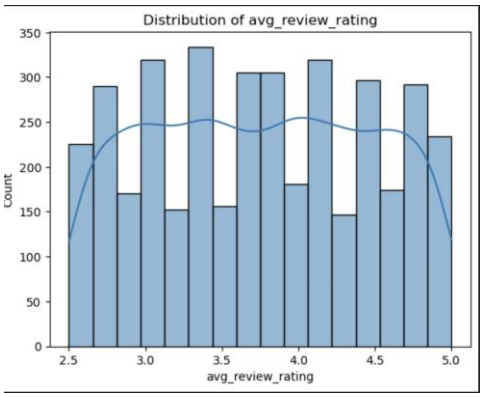


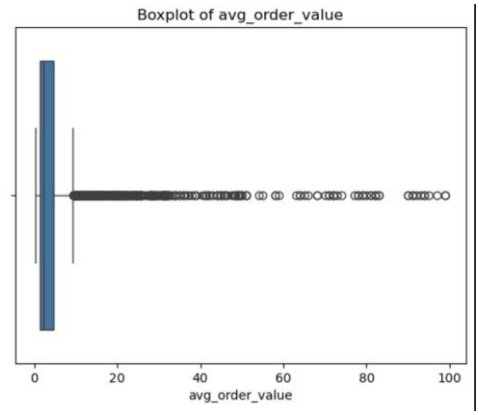

Confusion Matrix
Actual
Predicted
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
280
260
240
220
200
180
160
140

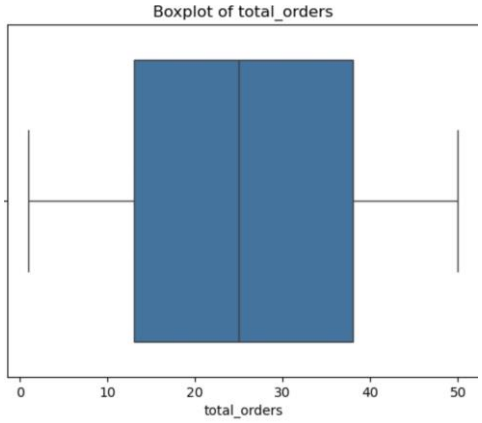

True Positive Rate (Positive label: 1)
False Positive Rate (Positive label: 1)
RandomForestClassifier (AUC = 0.52)
0.0
0.2
0.4
0.6
0.8
1.0

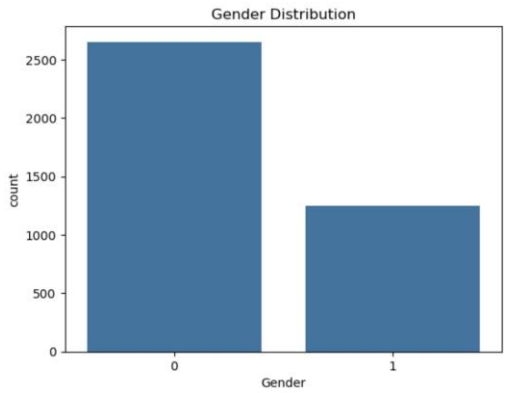

Distribution of avg_order_value
Count
avg_order_value
0
200
400
600
800
0
20
40
60
80
100

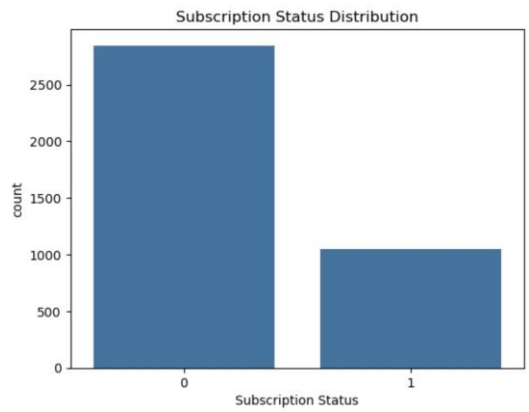

Distribution of total_orders
Count
total_orders
0
50
100
150
200
250
300
350
0
10
20
30
40
50

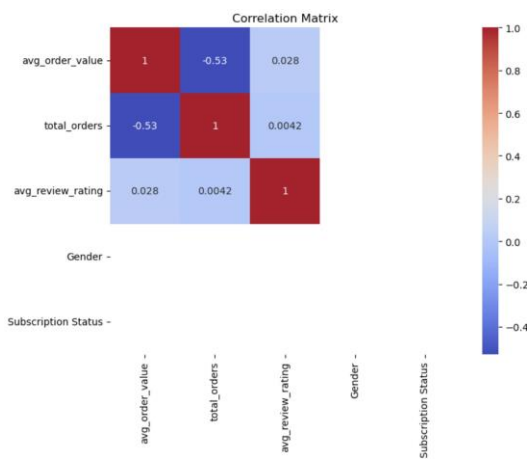

Distribution of avg_review_rating
Count
avg_review_rating
0
50
100
150
200
250
300
350
2.5
3.0
3.5
4.0
4.5
5.0


Boxplot of avg_order_value
avg_order_value
0
20
40
60
80
100


Boxplot of total_orders
total_orders
0
10
20
30
40
50

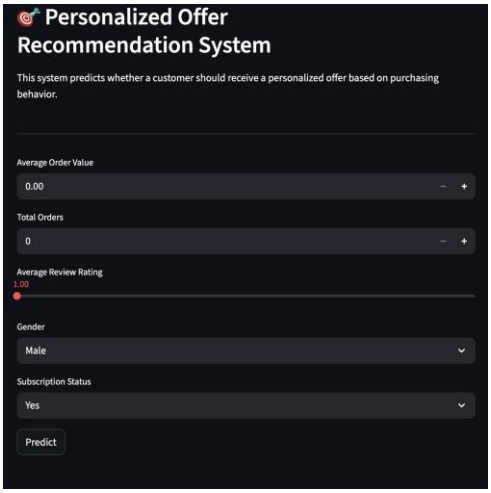


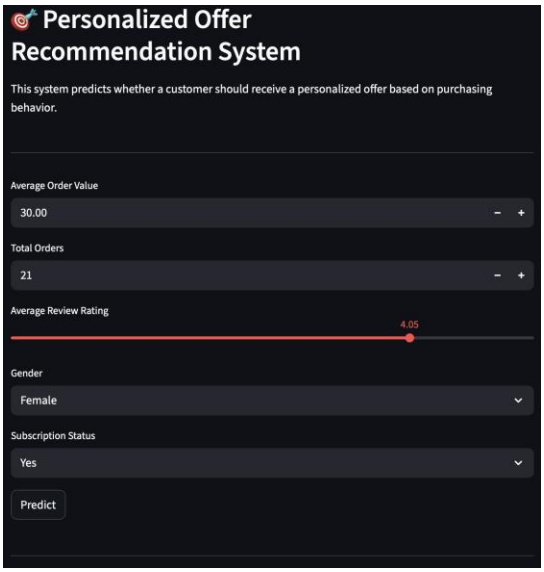


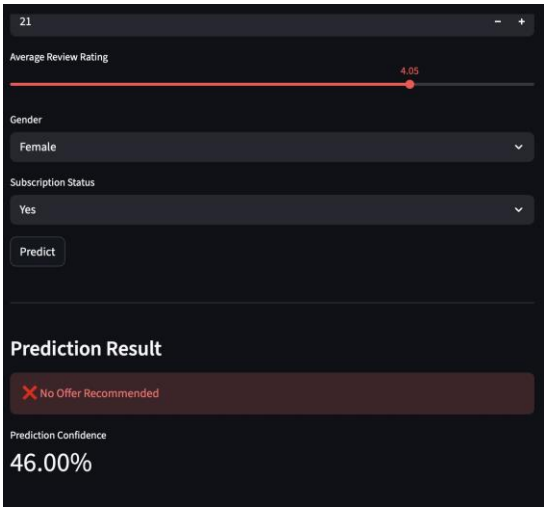


Results & Observations

- The model successfully predicts customer response to offers
- Explainability provides clear reasoning behind predictions
- Personalized recommendations can significantly improve engagement


🎯 Personalized Offer Recommendation System
This system predicts whether a customer should receive a personalized offer based on purchasing behavior.
Average Order Value
0.00
Total Orders
0
Average Review Rating
1.00
Gender
Male
Subscription Status
Yes
Predict


🎯 Personalized Offer Recommendation System
This system predicts whether a customer should receive a personalized offer based on purchasing behavior.
Average Order Value
30.00
Total Orders
21
Average Review Rating
4.05
Gender
Female
Subscription Status
Yes
Predict


21
Average Review Rating
4.05
Gender
Female
Subscription Status
Yes
Predict
Prediction Result
No Offer Recommended
Prediction Confidence
46.00%

Conclusion

In this project, an effective application of machine learning is demonstrated in order to provide personalized offers to customers. This is achieved by incorporating an explanation mechanism via SHAP, which provides transparency and trust in decision-making processes.

The proposed technique is useful in improving marketing strategies, conversion rates, and understanding consumer behaviour for retail businesses.

Future Enhancements

- Integration with real-time data sources (APIs)
- Advanced models like XGBoost or Deep Learning
- Deployment using Streamlit or Flask
- Real-time dashboards for business users

Tools & Technologies Used

- Python
- Pandas
- Scikit-learn
- SHAP
- Matplotlib / Seaborn (for visualization)
- Jupyter Notebook